

Dot products and weighted sums

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can calculate a dot product by matching feature positions.

I can explain how a weight changes a score with other inputs fixed.

I can distinguish a weighted score from a probability or trained model.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Representing inputs as vectors. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Calculate a dot product by matching feature positions

Explain how a weight changes a score with other inputs fixed

Distinguish a weighted score from a probability or trained model

Retrieve what you already know

Under schema [width, height], why must [4, 2] and [2, 4] remain distinct?

My starting explanation

This week's end goal

By the end, I can explain dot products and weighted sums using a traced example and a stated limitation.

2. Notice the evidence

A small model can combine several measurements into one score. Let $x = [4, 2]$ contain fictional width and height measurements in cm. Set weights $w = [2, -1]$. Multiply matching positions, then add: $s = (2 \text{ times } 4) + (-1 \text{ times } 2) = 6$. The letter x names the input vector.

We call this calculation a dot product, written $w \cdot x$. The vectors must have the same number of components. Each weight belongs to one named feature. These hand-selected weights demonstrate a computation; no training has happened. The result 6 is a score, not a probability or a claim that this parcel is safe.

Feature · Weight times input · Contribution

Feature: Width

Weight times input: 2 times 4

Contribution: 8

Feature: Height

Weight times input: -1 times 2

Contribution: -2

Feature: Total

Weight times input: 8 plus -2

Contribution: 6

The coefficients carry score-per-cm units in this example, making both contributions compatible score units. A weighted sum needs a consistent definition of units.

List the given facts and one fact that is not supplied.

3. See how it works

Keep $w = [2, -1]$ fixed. For $A = [4, 2]$, the score is 6. For $B = [2, 4]$, it is $4 - 4 = 0$. For $C = [3, 3]$, it is $6 - 3 = 3$. Swapping input positions without swapping their matching weights changes the computation. A zero score can come from cancellation, even when neither input is zero.

```
def dot(weights, inputs):  
    if len(weights) != len(inputs):  
        raise ValueError("Mismatched dimensions")  
    return sum(w * x for w, x in zip(weights, inputs))  
  
for row in [[4, 2], [2, 4], [3, 3]]:  
    print(dot([2, -1], row)) # 6, 0, 3
```

The length check matters: Python zip alone stops at the shorter list and could silently omit a feature. The function assumes numeric inputs already validated under the schema. First predict the three printed values on paper, then compare.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

For $w = [2, -1]$ and $x = [2, 4]$, explain why score zero does not mean the input contains no information.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Increase only the right input from 5 to 6.

Prediction: The score becomes $6 - 6 = 0$.

Reason: The right weight is -1, so adding 1 to that input lowers the score by 1.

5. Words that unlock the idea

Word · Meaning

Word: Weight

Meaning: A numerical coefficient multiplying one input component

Word: Dot product

Meaning: The sum of products of corresponding vector components

Word: Contribution

Meaning: The product of one input component and its matching weight

Word: Score

Meaning: A numerical output whose interpretation must be specified

Use two terms to explain a step in this week's example.

6. Follow a worked example

Multiply matching components then add

Position · Contribution rule · Worked contribution

Position: 1 width

Contribution rule: 2 times width

Worked contribution: 2 times 4 gives 8

Position: 2 height

Contribution rule: -1 times height

Worked contribution: -1 times 2 gives -2

Position: Add both

Contribution rule: 8 plus -2

Worked contribution: Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Keep $w = [2, -1]$. Change $[4, 2]$ to $[4, 3]$. Compute both contributions and explain the score change.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Dot products and weighted sums

Draw each input's contribution to a toy weighted brightness score.

The facts we will use

Inputs $[2, 5]$; weights $[3, -1]$; no bias. Score is $2 \times 3 + 5 \times (-1)$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Weighted brightness score

Input 2 x weight 3 = 6

Input 5 x weight -1 = -5

Add the products and bias 0: total = 1

Keep the total (identity) -> output 1

Dot products and weighted sums

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Multiply first input

Left contribution= $2 \times 3=6$.



2. Multiply second input

Right contribution= $5 \times (-1)=-5$.



3. Add contributions

Weighted sum= $6-5=1$; this is a score, not a probability.

CHANGE ONE THING

Increase only the right input from 5 to 6.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Multiply first input

What happens and why: Left contribution= $2 \times 3=6$.

Step: 2. Multiply second input

What happens and why: Right contribution= $5 \times (-1)=-5$.

Step: 3. Add contributions

What happens and why: Weighted sum= $6-5=1$; this is a score, not a probability.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A weight controls how strongly an input contributes to the score. The left connection adds three times its measurement; the negative right connection subtracts its measurement. Calculate each contribution before adding them. This dot product is a signed score; it becomes a probability only if a separate, justified transformation and interpretation are supplied.

What this example does not establish

These declared weights are hand-chosen toy parameters; no network training or calibrated probability is claimed.

7. Find and repair a mistake

Holding other inputs fixed, raising width by 1 cm raises this score by 2; raising height by 1 cm lowers it by 1. A negative weight reverses the direction of that feature contribution. It does not label the feature as bad, and its sign alone does not explain cause in the world.

Unit changes affect coefficients. If height is written in mm instead of cm while width remains in cm, use height weight -0.1 instead of -1 to preserve the same scores. For [4 cm, 20 mm], 2 times 4 plus -0.1 times 20 still gives 6. Raw coefficient magnitudes cannot establish feature importance when units differ.

Locate the earliest unsupported step

Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised weights and score.

State one repair and the checks you would rerun.

8. Practise together

For $w = [2, -1]$ and $x = [3, 5]$, write both contributions and their sum.

Keep $w = [2, -1]$. Change $[4, 2]$ to $[4, 3]$. Compute both contributions and explain the score change.

Explain your prediction to a partner before comparing results.

Practise with less help: Explain cancellation

Use workbook W2 for the connected example “Dot products and weighted sums”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Dot products and weighted sums. Proposed explanation: “Multiply the sums instead”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which connection explains why more right brightness lowers this score? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

For $w = [2, -1]$ and $x = [2, 4]$, explain why score zero does not mean the input contains no information.

```
def dot(weights, inputs):  
    if len(weights) != len(inputs):  
        raise ValueError("Mismatched dimensions")  
    return sum(w * x for w, x in zip(weights, inputs))  
  
for row in [[4, 2], [2, 4], [3, 3]]:  
    print(dot([2, -1], row)) # 6, 0, 3
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For $w = [2, -1]$ and $x = [3, 5]$, write both contributions and their sum.

Width stays in cm and height changes to mm. For the worked parcel [4 cm, 20 mm], give the revised weights and score.

Verify that [4, 2] and [5, 4] both score 6 with $w = [2, -1]$. Can a later step receiving only this score distinguish those two records?

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled dot products and weighted sums evidence sheet to Image-pattern Network Investigation.

Show how you can calculate a dot product by matching feature positions. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

A weighted sum can combine evidence, but it discards information: $[4, 2]$ and $[5, 4]$ both produce score 6 under these weights. A downstream rule receiving only that score cannot recover which of these vectors was used. To classify, we would also need a stated decision rule and labelled evaluation examples.

A teammate calls score 6 a 60 percent chance of class 1. What is missing from that claim?

Next week adds an offset called a bias and states exactly how a score becomes a class prediction.

Verify that [4, 2] and [5, 4] both score 6 with $w = [2, -1]$. Can a later step receiving only this score distinguish those two records?

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Multiply matching components then add

Position · Contribution rule · Worked contribution

Position: 1 width

Contribution rule: 2 times width

Worked contribution: 2 times 4 gives 8

Position: 2 height

Contribution rule: -1 times height

Worked contribution: -1 times 2 gives -2

Position: Add both

Contribution rule: 8 plus -2

Worked contribution: Score 6

The first two rows are separate contributions. The last row combines them; multiplying the final input sum by the final weight sum would be a different operation.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Compute a new dot product independently

Use $x = [3, 1, 2]$ and $w = [-2, 4, 0.5]$. Give each contribution and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-weight list is invalid for this input.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use $x = [3, 1, 2]$ and $w = [-2, 4, 0.5]$. Give each contribution and the total. Increase only the second component from 1 to 2 and compute the new total. Explain why a two-weight list is invalid for this input.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.